

Estate Planning Module - Complete Technical Documentation

Version: 1.0 **Last Updated:** 3 February 2026 **Module:** Estate Planning / IHT

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1. Overview

The Estate Planning module calculates Inheritance Tax (IHT) liability, generates mitigation strategies, and provides comprehensive estate planning recommendations. It integrates data from all other modules (Properties, Investments, Savings, Pensions, Protection) to build a complete picture of a user's estate.

Key Capabilities

- Real-time IHT liability calculation (current and projected at death)
- Spouse integration with combined estate calculations
- Gifting strategy optimisation (PETs, Annual Exemptions, Normal Expenditure)
- Trust strategy recommendations (CLTs, Loan Trusts, Discounted Gift Trusts)
- Life insurance coverage recommendations
- Will and bequest management
- Support for single, married, widowed users

Key Files

Component	Path
Agent	app/Agents/EstateAgent.php
IHT Calculation	app/Services/Estate/IHTCalculationService.php
Asset Aggregation	app/Services/Estate/EstateAssetAggregatorService.php

Component	Path
Gifting Strategy	app/Services/Estate/GiftingStrategyOptimizer.php
Trust Strategy	app/Services/Estate/PersonalizedTrustStrategyService.php
Comprehensive Plan	app/Services/Estate/ComprehensiveEstatePlanService.php
Controller	app/Http/Controllers/Api/EstateController.php
Vuex Store	resources/js/store/modules/estate.js
API Service	resources/js/services/estateService.js

2. Data Models

All estate models are located in `app/Models/Estate/`.

2.1 IHTProfile

Stores user's IHT planning profile and transferred allowances.

```
// app/Models/Estate/IHTProfile.php
protected $fillable = [
    'user_id',
    'marital_status',          // single, married, widowed
    'has_spouse',              // Boolean
    'own_home',                // Boolean – required for RNRB
    'home_value',              // Main residence value
    'nrb_transferred_from_spouse', // Transferred NRB from deceased spouse
    (widowed users)
    'charitable_giving_percent', // For reduced IHT rate (36% if 10%+ to
    charity)
];
```

2.2 Will

Stores will information and distribution instructions.

```
// app/Models/Estate/Will.php
protected $fillable = [
    'user_id',
    'has_will',                // Boolean
    'spouse_primary_beneficiary', // Boolean – spouse gets estate first
    'spouse_bequest_percentage', // 0–100% to spouse
    'executor_name',
    'executor_notes',
    'will_last_updated',       // Date
];

// Relationships
public function bequests(): HasMany // Individual bequests
```

2.3 Bequest

Individual bequests within a will.

```
// app/Models/Estate/Bequest.php
protected $fillable = [
    'will_id',
    'beneficiary_name',
    'beneficiary_user_id',      // If beneficiary is a system user
    'bequest_type',            // percentage_of_estate, specific_amount,
    specific_asset              // For percentage type
    'percentage_of_estate',    // For amount type
    'specific_amount',         // For intestacy fallback ordering
    'priority_order',
];
```

2.4 Gift

Records lifetime gifts for IHT calculation.

```
// app/Models/Estate/Gift.php
protected $fillable = [
    'user_id',
    'gift_date',
    'recipient',
    'gift_type',                // pet (Potentially Exempt Transfer), clt
    (Chargeable Lifetime Transfer), exempt
    'gift_value',
    'status',                   // pending, completed
    'taper_relief_applicable',  // Boolean - if 3-7 years since gift
];
```

2.5 Trust

Trust arrangements for IHT planning.

```
// app/Models/Estate/Trust.php
protected $fillable = [
    'user_id',
    'household_id',
    'trust_name',
    'trust_type',                // bare, discounted_gift, loan,
    life_insurance,              // interest_in_possession, discretionary,
    accumulation_maintenance
    'current_value',
    'loan_amount',               // For loan trusts
    'is_relevant_property_trust', // Subject to 10-year charges
];
```

```

        'last_periodic_charge_date',
        'beneficiaries',           // JSON array
        'trustees',               // JSON array
        'settlor',
    ];

```

2.6 Asset

Manual estate assets (separate from module-specific assets).

```

// app/Models/Estate/Asset.php
protected $fillable = [
    'user_id',
    'asset_type',           // property, investment, cash, business,
    chattel, etc.
    'asset_name',
    'current_value',
    'is_iht_exempt',       // Boolean
    'exemption_reason',    // BPR, charity, etc.
    'is_main_residence',   // Boolean – for RNRB
    'ownership_type',      // individual, joint, tenants_in_common,
    trust
    'beneficiary_designation', // Nominated beneficiary
];

```

2.7 Liability

Estate liabilities.

```

// app/Models/Estate/Liability.php
protected $fillable = [
    'user_id',
    'joint_owner_id',
    'liability_type',       // mortgage, loan, credit_card, etc.
    'liability_name',
    'current_balance',
    'monthly_payment',
    'interest_rate',
    'maturity_date',
    'is_priority_debt',     // Boolean
    'ownership_type',       // individual, joint
    'ownership_percentage', // Primary owner's share
];

```

2.8 IHTCalculation

Cached IHT calculation results.

```
// app/Models/Estate/IHTCalculation.php
protected $fillable = [
    'user_id',
    'user_gross_assets',
    'spouse_gross_assets',
    'total_gross_assets',
    'user_total_liabilities',
    'spouse_total_liabilities',
    'total_liabilities',
    'nrb_available',
    'nrb_message',
    'nrnb_available',
    'nrnb_status', // none, full, tapered
    'nrnb_message',
    'total_allowances',
    'taxable_estate',
    'iht_liability',
    'effective_rate',
    'projected_gross_assets',
    'projected_liabilities',
    'projected_net_estate',
    'projected_taxable_estate',
    'projected_iht_liability',
    'years_to_death',
    'estimated_age_at_death',
    'is_married',
    'data_sharing_enabled',
    'assets_hash', // For cache invalidation
    'liabilities_hash',
    'calculation_date',
];
```

3. Tax Configuration

CRITICAL: All IHT values come from [TaxConfigService](#) - NOTHING is hardcoded in calculation services.

The single source of truth is the [TaxConfiguration](#) database table, accessed via [TaxConfigService](#). This ensures consistency across all calculations and allows values to be updated without code changes.

3.1 Service Access Methods

```
// app/Services/TaxConfigService.php

// Core IHT configuration
$ihtConfig = $taxConfig->getInheritanceTax();

// Gifting exemptions
$giftingConfig = $taxConfig->getGiftingExemptions();

// PET (Potentially Exempt Transfer) rules
```

```
$petRules = $taxConfig->getPETRules();

// CLT (Chargeable Lifetime Transfer) rules
$cLtRules = $taxConfig->getCLTRules();

// 14-year rule configuration
$fourteenYearRule = $taxConfig->getFourteenYearRule();

// Trust charges (entry, periodic, exit)
$trustCharges = $taxConfig->getTrustCharges();

// Get taper relief for specific gift type
$taperRelief = $taxConfig->getTaperRelief('pet'); // or 'clt'

// Get tax rate based on years survived
$rate = $taxConfig->getGiftTaxRate(4.5, 'pet'); // Returns 0.24 (60% of 40%)

// Business/Agricultural Relief
$businessRelief = $taxConfig->getBusinessRelief();
$agriculturalRelief = $taxConfig->getAgriculturalRelief();

// Normal Expenditure from Income
$normalExpenditure = $taxConfig->getNormalExpenditureFromIncome();
```

3.2 Core Thresholds and Rates

All values from inheritance_tax.* in TaxConfigService:

Config Key	Value	Description
nil_rate_band	£325,000	Per person, frozen until April 2030
residence_nil_rate_band	£175,000	For main residence left to direct descendants
rnrB_taper_threshold	£2,000,000	RNRB tapers if estate exceeds this
rnrB_taper_rate	0.5	£1 lost per £2 over threshold
standard_rate	0.40	40% on taxable estate
reduced_rate_charity	0.36	36% if 10%+ left to charity
charity_threshold_percent	0.10	10% of baseline for reduced rate
spouse_exemption	true	Unlimited to UK-domiciled spouse
transferable_nil_rate_band	true	Unused NRB transfers to survivor
transferable_rnrB	true	Unused RNRB also transfers

3.3 Potentially Exempt Transfers (PETs)

Config path: inheritance_tax.potentially_exempt_transfers

Gifts to individuals that become exempt if donor survives 7 years.

Config Key	Value	Description
<code>years_to_exemption</code>	7	Fully exempt after 7 years
<code>immediate_charge</code>	false	No tax on gift when made
<code>becomes_chargeable_on_death</code>	true	Chargeable if donor dies within 7 years
<code>uses_donor_nrb</code>	true	Uses donor's NRB when calculating tax
<code>cumulation_period</code>	7	PETs in 7 years before death are cumulated

Taper Relief (`inheritance_tax.potentially_exempt_transfers.taper_relief`):

Years Survived	Tax Rate	Description
0-3	40%	Full rate
3-4	32%	80% of 40%
4-5	24%	60% of 40%
5-6	16%	40% of 40%
6-7	8%	20% of 40%
7+	0%	Fully exempt

Failed PET Rules (`inheritance_tax.potentially_exempt_transfers.failed_pet_rules`):

Rule	Value
<code>becomes_chargeable_transfer</code>	true
<code>affects_later_clt_nrb</code>	true
<code>affects_estate_nrb</code>	true
<code>calculation_order</code>	chronological

3.4 Chargeable Lifetime Transfers (CLTs)

Config path: `inheritance_tax.chargeable_lifetime_transfers`

Gifts to most trusts - immediately chargeable.

Config Key	Value	Description
<code>lookback_period</code>	7	CLTs in 7 years before this CLT use up NRB
<code>cumulation_period</code>	7	Rolling 7-year cumulation
<code>lifetime_rate</code>	0.20	20% immediate charge on excess over NRB
<code>lifetime_rate_grossed_up</code>	0.25	25% if settlor pays the tax
<code>death_rate</code>	0.40	40% rate on death within 7 years
<code>additional_death_charge</code>	0.20	Extra 20% due if death within 7 years

Config Key	Value	Description
<code>taper_relief_applies</code>	true	Taper relief reduces additional charge

CLT Taper Relief (`inheritance_tax.chargeable_lifetime_transfers.taper_relief`):

Years	Relief %	Effective Tax %
0-3	0%	100% of additional charge
3-4	20%	80%
4-5	40%	60%
5-6	60%	40%
6-7	80%	20%
7+	100%	0% (no additional charge)

3.5 The 14-Year Rule

Config path: `inheritance_tax.fourteen_year_rule`

When calculating IHT on a CLT made within 7 years of death, failed PETs made in the 7 years before that CLT reduce the available NRB.

Config Key	Value	Description
<code>applies_to</code>	<code>clt_with_prior_failed_pet</code>	Specific scenario
<code>lookback_for_failed_pets</code>	7	Years to look back for failed PETs
<code>lookback_for_clts</code>	7	Years before death to check CLTs
<code>maximum_window</code>	14	Total maximum window

Calculation Steps (from config):

1. Identify all CLTs made within 7 years of death
2. For each CLT, identify any failed PETs made in the 7 years before that CLT
3. Failed PETs reduce the NRB available for the CLT
4. This can result in additional tax on the CLT even if it was within NRB when made

3.6 Trust Charges

Config path: `inheritance_tax.trust_charges`

Entry Charge

Config Key	Value	Description
<code>entry.rate</code>	0.20	20% on value exceeding NRB
<code>entry.rate_grossed_up</code>	0.25	25% if settlor pays

Config Key	Value	Description
<code>entry.nrb_available</code>	true	NRB applies (less previous 7-year CLTs)

Periodic (10-Year) Charge

Config Key	Value	Description
<code>periodic.interval_years</code>	10	Charged every 10 years
<code>periodic.max_rate</code>	0.06	Maximum 6% of trust value
<code>periodic.calculation_formula</code>	"30% of lifetime rate"	30% × 20% = 6%
<code>periodic.lifetime_rate_multiplier</code>	0.30	30% multiplier
<code>periodic.base_rate</code>	0.20	Base lifetime rate
<code>periodic.nrb_available</code>	true	NRB reduces the charge

Formula: Trust Value × Effective Rate

- Effective Rate = 30% × (Excess over NRB / Trust Value) × 20%
- Maximum effective rate = 6%

Exit Charge

Config Key	Value	Description
<code>exit.max_rate</code>	0.06	Maximum 6%
<code>exit.calculation_basis</code>	proportionate	Based on time since last anniversary
<code>exit.quarters_in_period</code>	40	40 complete quarters in 10 years
<code>exit.formula</code>	"effective_rate × 30% × (quarters / 40)"	Proportionate formula

No Exit Charge Periods:

Scenario	Period
After trust setup	3 months
After 10-year anniversary	3 months
Discretionary will trust	24 months from death

3.7 Gifting Exemptions

Config path: `gifting_exemptions`

Annual Exemption

Config Key	Value
annual_exemption	£3,000
annual_exemption_can_carry_forward	true
carry_forward_years	1

Small Gifts

Config Key	Value
small_gifts_limit	£250 per recipient
small_gifts_unlimited_recipients	true

Wedding/Civil Partnership Gifts

Relationship	Amount
parent_to_child	£5,000
grandparent_to_grandchild	£2,500
other_person	£1,000

Normal Expenditure from Income

Config path: `gifting_exemptions.normal_expenditure_from_income`

Config Key	Value
limit	null (unlimited)
immediately_exempt	true

Conditions (all must be met):

- From income, not capital
- Regular/habitual pattern
- Does not affect donor's standard of living

Evidence Required:

- Income and expenditure records
- Pattern of regular giving (typically 3+ years)
- Proof that standard of living maintained

3.8 Business & Agricultural Relief

Config path: `inheritance_tax.business_relief` and `inheritance_tax.agricultural_relief`

Asset Type	Relief Rate
Trading business (sole trader/partnership)	100%

Asset Type	Relief Rate
Unquoted company shares	100%
AIM-listed shares	100%
Controlling holding in quoted company	100%
Land/buildings used by controlled business	50%
Investment companies	0%

Requirements:

- Minimum 2 years ownership
- Trading status (not investment holding)
- From April 2026: Combined APR/BPR capped at £1m at 100%, then 50%

3.9 Quick Succession Relief

Config path: `inheritance_tax.quick_succession_relief`

Applies when beneficiary dies shortly after receiving an inheritance.

Years After Receiving	Relief
0-1	100%
1-2	80%
2-3	60%
3-4	40%
4-5	20%

3.10 Assumptions (for Projections)

Config path: `assumptions`

Estate projections use sophisticated asset-specific growth models, all configurable via **Settings > Assumptions > Estate Planning**.

Asset-Specific Growth Methods

Unlike simple flat-rate growth, estate projections apply different growth methodologies to different asset types:

Asset Type	Growth Method	Configurable	Default
Cash Accounts	Income/Expense Surplus Model	No	See below
Investment Accounts	Monte Carlo (80% confidence)	Yes	Monte Carlo
Properties	Fixed Annual %	Yes	3%
Liabilities	Amortisation to end date	No	End at retirement

Cash Account Projection (Income/Expense Surplus Model)

Cash accounts do **NOT** use a flat growth rate. Instead, the projection models the user's actual cash flow:

Pre-Retirement (current age → retirement age):

Annual Surplus = Total Annual Income – Total Annual Expenses

Where:

Total Annual Income = user.annual_employment_income
 + user.annual_self_employment_income
 + user.annual_rental_income
 + user.annual_dividend_income
 + user.annual_interest_income
 + user.annual_other_income
 + user.annual_trust_income
 + spouse income (if linked and data sharing enabled)

Total Annual Expenses = expenditure_profile.total_monthly_expenditure × 12

Cash grows by adding annual surplus each year (can be negative if deficit).

Post-Retirement (retirement age → death):

Annual Surplus = Retirement Income – Retirement Expenses

Where:

Retirement Income = retirement_profile.target_retirement_income
 + State Pension (from state_pension_age onwards)
 + Pension income (DC/DB drawdown)
 + spouse retirement income (if linked)

Retirement Expenses = retirement_profile.essential_expenditure
 + retirement_profile.lifestyle_expenditure

State Pension timing:

- Included from user.state_pension_age (typically 66–68)
- Amount from state_pensions.estimated_annual_amount
- For couples, both state pensions included at respective ages

Investment Growth (Monte Carlo 80% Confidence)

Default Method: Monte Carlo Simulation

- Uses the 20th percentile (p20) from 1000 Monte Carlo iterations
- This represents the value that has an 80% probability of being achieved or exceeded
- Calculated by [MonteCarloSimulator](#) using normal distribution (Box-Muller transform)
- Risk-adjusted based on portfolio allocation
- Already used in the Net Worth module's Investments tab

Alternative: Custom Rate

- User specifies a fixed annual growth percentage
- Applied as simple compound growth: $FV = PV \times (1 + r)^n$
- Useful for conservative or aggressive projections

Property Growth (3% Default)

- All properties grow at a fixed annual rate (default 3%)
- Configurable per user via Assumptions settings
- Same rate applied to main residence, secondary residences, and buy-to-let

Liability Projection

Liabilities are projected to their end date:

Liability Type	End Date Logic
Mortgages	Use <code>mortgage.end_date</code> if set, otherwise assume retirement age
Other Liabilities	Use <code>liability.end_date</code> if set, otherwise assume retirement age
No end date	Default to <code>retirement_profile.target_retirement_age</code> or <code>user.target_retirement_age</code>

Liabilities reduce linearly to £0 at their end date (simplified amortisation).

User-Configurable Options

The Assumptions Settings page (`/settings/assumptions`) allows users to customise:

Setting	Options	Default
Inflation Rate	0% - 20% (editable)	2.5%
Property Growth Rate	-10% - 20% (editable)	3%
Investment Growth Method	Dropdown: "Monte Carlo (80%)" or "Custom %"	Monte Carlo (80%)
Custom Investment Rate	-10% to 30% (only shown if Custom selected)	—

Note: Cash account growth is NOT configurable as it's derived from actual income/expenses.

Data Sources

PROJECTION DATA SOURCES	
INCOME (Pre-Retirement)	INCOME (Post-Retirement)
<code>users.annual_employment_income</code>	<code>retirement_profiles.</code>

users.annual_self_employment_income	target_retirement_income
users.annual_rental_income	state_pensions.
users.annual_dividend_income	estimated_annual_amount
users.annual_interest_income	(from state_pension_age)
users.annual_other_income	
users.annual_trust_income	
<u>EXPENSES (Pre-Retirement)</u>	<u>EXPENSES (Post-Retirement)</u>
expenditure_profiles.	retirement_profiles.
total_monthly_expenditure × 12	essential_expenditure
	retirement_profiles.
	lifestyle_expenditure
<u>RETIREMENT AGE</u>	
retirement_profiles.target_retirement_age	
OR users.target_retirement_age	
OR dc_pensions.retirement_age (fallback)	
OR 68 (default)	

How Values Flow

User Settings (UserAssumption model)
↓
AssumptionsService.getEstateAssumptions()
↓
IHTCalculationService.calculateProjectedValues()
↓

For each year from current_age to estimated_death_age:

CASH ACCOUNTS:
if year < retirement_age:
cash += (total_income - current_expenses)
else:
retirement_income = target_retirement_income
if year >= state_pension_age:
retirement_income += state_pension_amount
cash += (retirement_income - retirement_expenses)

INVESTMENTS:
if monte_carlo_method:
investments = MonteCarloSimulator.getP20Value(years)
else:
investments *= (1 + custom_rate)

PROPERTIES:
properties *= (1 + property_growth_rate)

LIABILITIES:

```

    if year <= liability_end_year:
        remaining_term = liability_end_year - year
        liability = original * (remaining_term / total_term)
    else:
        liability = 0

```

↓

Projected Estate at Death = Cash + Investments + Properties - Liabilities

Storage

User overrides are stored in `user_assumptions` table:

```

UserAssumption::where('assumption_type', 'estate_planning')
    ->where('user_id', $userId)
    ->first();

// Fields:
// - inflation_rate (decimal)
// - property_growth_rate (decimal)
// - investment_growth_method (enum: 'monte_carlo', 'custom')
// - custom_investment_rate (decimal, nullable)

```

If no override exists, defaults from `TaxConfigService.getAssumptions()` are used.

4. Backend Architecture

4.1 EstateAgent

The orchestrator that coordinates all estate planning calculations.

File: `app/Agents/EstateAgent.php`

```

class EstateAgent extends BaseAgent
{
    public function __construct(
        private IHTCalculationService $ihtCalculator,
        private EstateAssetAggregatorService $assetAggregator,
        private ComprehensiveEstatePlanService $estatePlanService,
        private GiftingStrategyOptimizer $giftingOptimizer,
        private PersonalizedTrustStrategyService $trustStrategyService
    ) {}
}

```

Key Methods

`analyze($userId)` (line 34-120)

- Loads user with all relationships
- Aggregates estate assets via `EstateAssetAggregatorService`
- Calculates IHT via `IHTCalculationService`
- Calculates estate health score (0-100)
- Gets trust recommendations via `PersonalizedTrustStrategyService`
- Gets gifting opportunities via `GiftingStrategyOptimizer`

`generateRecommendations($analysisData)` (line 125-204)

IHT Mitigation Decision Tree:

IF IHT Liability > £0:

- STEP 1: CHARITABLE BEQUEST CHECK (Rate Reduction)
 - Check user.charitable_bequest toggle
 - IF TRUE: Calculate baseline (Net Estate – NRB)
 - Minimum donation = baseline × 10%
 - Apply reduced rate: 36% instead of 40%
 - IF FALSE: Use standard 40% rate
 - Recalculate IHT liability with effective rate
- STEP 2: LIQUIDITY & AFFORDABILITY CHECK
 - Assess which assets can be utilised (liquid vs illiquid)
 - Calculate available surplus income
 - Identify any affordability constraints
- STEP 3: CHECK EXISTING LIFE COVER
 - Query Protection module for existing policies
 - Check if policies are written in trust (bypasses estate)
 - Calculate excess cover above outstanding debts
 - Only usable cover = (Total Cover – Outstanding Debts)
- STEP 4: ANNUAL GIFTING STRATEGY (First Resort)
 - Calculate annual exemption capacity (£3,000 × years to death)
 - Add normal expenditure from income potential
 - Add small gifts allowance (£250 per recipient)
 - IF annual gifting covers IHT liability → RECOMMEND
 - IF NOT → Calculate remaining liability after gifting
- STEP 5: LIFE COVER STRATEGY (Second Resort)
 - IF user age ≤ 50:
 - Calculate whole of life premium for remaining liability
 - Check affordability against surplus income
 - RECOMMEND whole of life in trust if affordable
 - IF user age > 50:
 - Life cover may be prohibitively expensive
 - Skip to PET strategy if no existing cover
 - Only use existing cover excess for mitigation
 - Remaining liability = IHT – Gifting – Life Cover Excess
- STEP 6: PET GIFTING STRATEGY (Third Resort)
 - Calculate 7-year cycle capacity
 - IF user age > 50 AND no life cover:

```
├─ PRIORITISE PETs over new life cover
├─ Apply taper relief projections
├─ Calculate remaining liability after PETs
└─ STEP 7: CLT INTO TRUST (Last Resort Only)
    ├─ Only if Steps 4–6 do not fully cover liability
    ├─ Calculate immediate 20% charge
    ├─ Calculate potential additional 20% on death within 7 years
    ├─ Consider periodic (10-year) and exit charges
    └─ RECOMMEND with full cost/benefit analysis
```

buildScenarios(\$userId, \$parameters) (line 209-240)

- Builds what-if scenarios: current, optimized, gifting, property downsizing, trust creation
- Each scenario shows projected IHT liability, estate value, and beneficiary amounts

calculateEstateHealthScore(\$user, \$assetSummary, \$ihtLiability) (line 245-284)

```
Base score: 100
Deductions:
-20 Missing IHT profile
-25 IHT ratio > 30%
-15 IHT ratio > 20%
-10 IHT ratio > 10%
-10 Estate > £650k with no trusts
-5 Married without linked spouse
-15 IHT liability > 50% of liquid assets
```

4.2 Service Layer

Service	Purpose
IHTCalculationService	Core IHT calculation, NRB/RNRB, projections
EstateAssetAggregatorService	Aggregates assets from all modules
GiftingStrategyOptimizer	Calculates optimal gifting strategy
PersonalizedTrustStrategyService	Generates trust recommendations
ComprehensiveEstatePlanService	Builds complete estate plan document
AssetLiquidityAnalyzer	Analyzes asset liquidity for trust planning
NetWorthAnalyzer	Net worth calculations and breakdowns
FutureValueCalculator	Compound growth calculations

5. Asset Aggregation

5.1 Single-Record Architecture

The system uses a **single-record pattern** for joint assets:

- ONE database record stores the FULL asset value
- `user_id` = primary owner
- `joint_owner_id` = secondary owner (if joint)
- `ownership_percentage` = primary owner's share

File: `app/Services/Estate/EstateAssetAggregatorService.php`

```
// Query pattern for all modules
$assets = Model::where('user_id', $user->id)
    ->orWhere('joint_owner_id', $user->id)
    ->get();

// Ownership calculation
$userIsOwner = $asset->user_id === $user->id;
$share = $userIsOwner
    ? ($ownership_percentage / 100)
    : ((100 - $ownership_percentage) / 100);
$userValue = $asset->full_value * $share;
```

5.2 Asset Sources

```
EstateAssetAggregatorService.gatherUserAssets($user)
├─ Manual Estate Assets (Asset model)
│   └─ Direct user entries
├─ Properties (Property model)
│   ├── Query: user_id OR joint_owner_id
│   ├── Apply: ownership_percentage
│   └─ property_type: main_residence, secondary_residence, buy_to_let
├─ Investment Accounts (InvestmentAccount model)
│   ├── Query: user_id OR joint_owner_id
│   ├── Apply: ownership_percentage
│   └─ ISAs are NOT IHT-exempt
├─ Savings Accounts (SavingsAccount model)
│   ├── Query: user_id OR joint_owner_id
│   └─ Apply: ownership_percentage
├─ Business Interests (BusinessInterest model)
│   ├── Query: user_id OR joint_owner_id
│   ├── BPR eligibility check:
│   │   └─ bpr_eligible=true AND trading_status='trading' AND 2+ years owned
│   └─ 100% IHT exempt if qualifying
├─ Chattels (Chattel model)
│   ├── Query: user_id OR joint_owner_id
│   └─ Apply: ownership_percentage
├─ DC Pensions (DCPension model)
│   ├── is_iht_exempt = true
│   └─ Outside estate if beneficiary nominated
└─ DB Pensions (DBPension model)
    ├── current_value = 0
    └─ Die with member (no transfer value)
```

5.3 Liability Sources

```
EstateAssetAggregatorService.calculateUserLiabilities($user)
├─ Mortgages (Mortgage model)
│   └─ Query: user_id OR joint_owner_id
│       └─ Apply: ownership share
├─ Estate Liabilities (Liability model)
│   └─ Query: user_id OR joint_owner_id
│       └─ Apply: ownership share
```

5.4 Standardised Asset Object

All assets are converted to a standardised format:

```
(object) [
    'user_id' => $user->id,
    'asset_type' => 'investment',           // property, cash, business, etc.
    'asset_name' => 'Provider - ISA',
    'current_value' => 50000,               // User's share
    'full_value' => 100000,                // Total asset value
    'ownership_type' => 'joint',            // individual, joint,
    tenants_in_common
    'ownership_percentage' => 50,           // Primary owner's %
    'is_primary_owner' => true,
    'is_iht_exempt' => false,
    'property_type' => null,               // For RNRB: main_residence, etc.
    'bpr_eligible' => false,              // For business assets
];
```

6. IHT Calculation Logic

File: `app/Services/Estate/IHTCalculationService.php`

6.1 Calculation Flow

```
public function calculate(User $user, ?User $spouse = null, bool
$dataSharingEnabled = false): array
```

Step-by-step:

1. Check Cache

- Compare asset/liability hashes
- Return cached result if unchanged

2. Get Tax Config

- NRB, RNRB, rates from `TaxConfigService`

3. Aggregate Taxable Assets

- User assets via `EstateAssetAggregatorService`
- Spouse assets (if married + data sharing)
- Filter out IHT-exempt (DC pensions, BPR business)

4. Calculate Liabilities

- User liabilities
- Spouse liabilities (if applicable)

5. Calculate Net Estate

```
Net Estate = Gross Assets - Liabilities
```

6. Calculate NRB

```
Single: £325,000
Married: £650,000 (£325k × 2)
Widowed: Up to £650,000 (if spouse's NRB transferred)
```

7. Calculate RNRB (see 6.2)

8. Calculate Taxable Estate

```
Taxable = Net Estate - NRB - RNRB
```

9. Calculate IHT

```
IHT = Taxable Estate × 40%
```

10. Calculate Projections (see Section 7)

11. Save to Database

- Store in `IHTCalculation` table with hashes

6.2 RNRB Calculation

File: `IHTCalculationService.php` lines 222-275

```
private function calculateRNRB(
    float $totalNetEstate,
    User $user,
```

```
?User $spouse,
array $ihtConfig,
bool $isMarried
): array
```

Eligibility Requirements:

- Must own main residence (`property_type = 'main_residence'`)
- Must leave to direct descendants (children, grandchildren)

Calculation:

```
Full RNRB:
- Single: £175,000
- Married: £350,000 (£175k × 2)

If estate > £2,000,000:
  Excess = Estate - £2,000,000
  Reduction = Excess × 0.5 (£1 per £2)
  RNRB Available = Full RNRB - Reduction

If Reduction > Full RNRB:
  RNRB Available = £0 (fully tapered)
```

Status Values:

- `full` - Full RNRB available
- `tapered` - Partially or fully reduced
- `none` - No main residence

6.3 Result Structure

```
[
  // Current values
  'user_gross_assets' => 500000,
  'spouse_gross_assets' => 300000,
  'total_gross_assets' => 800000,

  'user_total_liabilities' => 150000,
  'spouse_total_liabilities' => 50000,
  'total_liabilities' => 200000,

  'user_net_estate' => 350000,
  'spouse_net_estate' => 250000,
  'total_net_estate' => 600000,

  'nrb_available' => 650000,
  'nrb_message' => 'Combined Nil Rate Band of £650,000...',

  'rnrn_available' => 350000,
  'rnrn_status' => 'full',
```

```

        'nrnb_message' => 'Full Residence Nil Rate Band of £350,000...',

        'total_allowances' => 1000000,
        'taxable_estate' => 0,
        'iht_liability' => 0,
        'effective_rate' => 0,

        // Projected values at death
        'projected_gross_assets' => 1200000,
        'projected_liabilities' => 200000,
        'projected_net_estate' => 1000000,
        'projected_taxable_estate' => 0,
        'projected_iht_liability' => 0,
        'years_to_death' => 25,
        'estimated_age_at_death' => 85,

        'is_married' => true,
        'data_sharing_enabled' => true,
    ];

```

7. Projection Calculations

File: `IHTCalculationService.php` lines 152-350

Estate projections calculate the expected estate value at the user's (or surviving spouse's) estimated death. Unlike simple flat-rate projections, the system uses asset-specific growth models that account for income, expenses, and lifecycle changes.

7.1 Life Expectancy

```

private function calculateLifeExpectancy(User $user): int
{
    // Query actuarial_life_tables
    $lifeExpectancy = DB::table('actuarial_life_tables')
        ->where('age', '<=', $currentAge)
        ->where('gender', $user->gender)
        ->where('table_year', '2020-2022')
        ->orderBy('age', 'desc')
        ->first();

    // Fallback: 85 - current_age
}

```

For Married Couples:

- Calculate life expectancy for BOTH spouses
- Use the LONGER life expectancy (second death scenario)
- IHT is calculated on the combined estate at second death

7.2 Retirement Age Determination

The retirement age is a critical transition point in projections:

```
private function getRetirementAge(User $user): int
{
    // Priority order for retirement age
    if ($user->retirementProfile?->target_retirement_age) {
        return $user->retirementProfile->target_retirement_age;
    }

    if ($user->target_retirement_age) {
        return $user->target_retirement_age;
    }

    // Check DC pensions for retirement age
    $pensionRetirementAge = $user->dcPensions()
        ->whereNotNull('retirement_age')
        ->value('retirement_age');

    if ($pensionRetirementAge) {
        return $pensionRetirementAge;
    }

    return 68; // Default UK state pension age
}
```

7.3 Asset-Specific Projection Methods

7.3.1 Cash Account Projection (Income/Expense Surplus Model)

Cash accounts do **NOT** use a flat growth percentage. Instead, they grow (or shrink) based on the user's income minus expenses.

```
private function projectCashAccounts(
    User $user,
    float $currentCash,
    int $currentAge,
    int $retirementAge,
    int $deathAge
): float {
    $projectedCash = $currentCash;

    // Pre-Retirement Phase: currentAge → retirementAge
    for ($age = $currentAge; $age < $retirementAge; $age++) {
        $annualIncome = $this->getTotalAnnualIncome($user);
        $annualExpenses = $this->getCurrentAnnualExpenses($user);
        $surplus = $annualIncome - $annualExpenses;
        $projectedCash += $surplus;
    }

    // Post-Retirement Phase: retirementAge → deathAge
    for ($age = $retirementAge; $age < $deathAge; $age++) {
```

```

        $retirementIncome = $this->getRetirementIncome($user, $age);
        $retirementExpenses = $this->getRetirementExpenses($user);
        $surplus = $retirementIncome - $retirementExpenses;
        $projectedCash += $surplus;
    }

    return max(0, $projectedCash); // Cannot go negative (would draw from
investments)
}

```

Income Calculation (Pre-Retirement):

```

private function getTotalAnnualIncome(User $user): float
{
    $income = $user->annual_employment_income
        + $user->annual_self_employment_income
        + $user->annual_rental_income
        + $user->annual_dividend_income
        + $user->annual_interest_income
        + $user->annual_other_income
        + $user->annual_trust_income;

    // Include spouse income if linked and data sharing enabled
    if ($user->spouse && $user->dataSharingEnabled()) {
        $income += $this->getTotalAnnualIncome($user->spouse);
    }

    return $income;
}

```

Expenses Calculation with Fallbacks:

```

// Pre-Retirement (from ExpenditureProfile with income-based fallback)
private function getCurrentAnnualExpenses(User $user): float
{
    $profile = $user->expenditureProfile;
    if ($profile && $profile->total_monthly_expenditure > 0) {
        return $profile->total_monthly_expenditure * 12;
    }

    // FALLBACK: Use 70% of total annual income as estimated expenses
    // This is more realistic than £0 and prevents unrealistic cash growth
    $totalIncome = $this->getTotalAnnualIncome($user);
    return $totalIncome * 0.70;
}

// Post-Retirement (from RetirementProfile with fallbacks)
private function getRetirementExpenses(User $user): float
{
    $profile = $user->retirementProfile;

```

```
// Priority 1: Use explicit retirement expenses if set
if ($profile) {
    $expenses = ($profile->essential_expenditure ?? 0) + ($profile->
>lifestyle_expenditure ?? 0);
    if ($expenses > 0) {
        return $expenses;
    }

    // Priority 2: Use target_retirement_income as proxy for expenses
    if ($profile->target_retirement_income > 0) {
        return $profile->target_retirement_income;
    }
}

// Priority 3: Use 50% of pre-retirement income as retirement expenses
// (assumes reduced lifestyle in retirement)
$preRetirementIncome = $this->getTotalAnnualIncome($user);
return $preRetirementIncome * 0.50;
}
```

Expense Fallback Summary:

Scenario	Pre-Retirement Expenses	Retirement Expenses
Full data	total_monthly_expenditure × 12	essential + lifestyle
No expenditure profile	income × 70%	target_retirement_income
No retirement profile	income × 70%	pre-retirement income × 50%
No income data	£0	£0

Retirement Income with State Pension:

```
private function getRetirementIncome(User $user, int $age): float
{
    $income = $user->retirementProfile?->target_retirement_income ?? 0;

    // Add State Pension if user has reached state pension age
    $statePensionAge = $user->state_pension_age ?? 67;
    if ($age >= $statePensionAge) {
        $statePension = $user->statePensions()->first();
        $income += $statePension?->estimated_annual_amount ?? 0;
    }

    // Include spouse retirement income and state pension
    if ($user->spouse && $user->dataSharingEnabled()) {
        $spouseIncome = $user->spouse->retirementProfile?->
>target_retirement_income ?? 0;
        $spouseStatePensionAge = $user->spouse->state_pension_age ?? 67;

        if ($age >= $spouseStatePensionAge) {
            $spouseStatePension = $user->spouse->statePensions()->first();
            $spouseIncome += $spouseStatePension?->estimated_annual_amount ??
```

```

0;
    }

    $income += $spouseIncome;
}

return $income;
}

```

7.3.2 Investment Account Projection (Monte Carlo 80%)

Investment accounts use Monte Carlo simulation by default, returning the 80% confidence value:

```

private function projectInvestments(
    User $user,
    int $yearsToProject
): float {
    $assumptions = $this->assumptionsService->getEstateAssumptions($user);

    if ($assumptions['investment_growth_method'] === 'monte_carlo') {
        // Monte Carlo: returns p20 (80% probability of achieving or
        exceeding)
        return $this->investmentProjectionService
            ->getAccountProjectedValue80($user, $yearsToProject);
    }

    // Custom rate: simple compound growth
    $customRate = $assumptions['custom_investment_rate'] / 100;
    $currentValue = $this->getTotalInvestmentValue($user);

    return $currentValue * pow(1 + $customRate, $yearsToProject);
}

```

Monte Carlo Implementation:

- 1000 iterations using Box-Muller transform for normal distribution
- Expected returns based on portfolio risk profile
- Returns 20th percentile (p20) = 80% confidence level
- Factors in annual volatility and mean reversion

7.3.3 Property Projection (3% Default, Configurable)

Properties use a fixed annual growth rate:

```

private function projectProperties(
    User $user,
    int $yearsToProject
): float {
    $assumptions = $this->assumptionsService->getEstateAssumptions($user);
    $propertyGrowthRate = $assumptions['property_growth_rate'] / 100; //

```

Default 3%

```

$currentPropertyValue = $user->properties->sum('current_value');

// Include spouse properties if linked
if ($user->spouse && $user->dataSharingEnabled()) {
    $currentPropertyValue += $user->spouse->properties-
>sum('current_value');
}

return $currentPropertyValue * pow(1 + $propertyGrowthRate,
$yearsToProject);
}

```

7.3.4 Liability Projection (Amortisation to End Date)

Liabilities reduce to zero at their end date. If no end date is specified, retirement age is assumed:

```

private function projectLiabilities(
    User $user,
    int $currentAge,
    int $retirementAge,
    int $deathAge
): float {
    $projectedLiabilities = 0;
    $yearsToProject = $deathAge - $currentAge;

    // Project mortgages
    foreach ($user->mortgages as $mortgage) {
        $endDate = $mortgage->end_date;

        // If no end date, assume paid off at retirement
        if (!$endDate) {
            $endYear = $retirementAge - $currentAge;
        } else {
            $endYear = Carbon::parse($endDate)->year - now()->year;
        }

        if ($yearsToProject < $endYear) {
            // Still have remaining balance at death
            $remainingTerm = $endYear - $yearsToProject;
            $totalTerm = $endYear;
            $projectedBalance = $mortgage->outstanding_balance *
($remainingTerm / $totalTerm);
            $projectedLiabilities += max(0, $projectedBalance);
        }
        // else: mortgage paid off before death, contributes £0
    }

    // Project other liabilities
    foreach ($user->liabilities as $liability) {
        $endDate = $liability->end_date;
    }
}

```

```

    // If no end date, assume cleared at retirement
    if (!$endDate) {
        $endYear = $retirementAge - $currentAge;
    } else {
        $endYear = Carbon::parse($endDate)->year - now()->year;
    }

    if ($yearsToProject < $endYear) {
        $remainingTerm = $endYear - $yearsToProject;
        $totalTerm = $endYear;
        $projectedBalance = $liability->current_balance * ($remainingTerm
/ $totalTerm);
        $projectedLiabilities += max(0, $projectedBalance);
    }
}

return $projectedLiabilities;
}

```

7.3.5 Integrated Cash-Investment Drawdown Model

When cash surplus turns negative during retirement, the deficit is drawn from investment accounts. This creates an integrated year-by-year projection where:

1. **Cash deficit reduces investments** - Before applying growth
2. **Deficit split equally** - Across all investment accounts
3. **Growth applied after drawdown** - Investment growth happens on reduced balance

```

/**
 * Integrated projection: Cash deficits drawn from investments year-by-year
 *
 * For each year:
 * 1. Calculate cash surplus (income - expenses)
 * 2. If negative, deduct deficit from investments BEFORE applying growth
 * 3. Apply investment growth rate to reduced balance
 * 4. Repeat for each year until death
 */
private function projectCashAndInvestmentsIntegrated(
    User $user,
    ?User $spouse,
    int $currentAge,
    int $retirementAge,
    int $deathAge,
    bool $dataSharingEnabled
): array {
    // Get initial values
    $cashBalance = $this->getCurrentCashValue($user);
    $investments = $this->getInvestmentAccounts($user, $spouse,
$dataSharingEnabled);
    $totalInvestments = array_sum(array_column($investments, 'balance'));

    // Get growth rate (Monte Carlo 80% annualised or custom)

```

```

    $assumptions = $this->assumptionsService->getEstateAssumptions($user);
    $investmentGrowthRate = $assumptions['investment_growth_method'] ===
'monte_carlo'
    ? $this->getMonteCarloAnnualRate($user)
    : ($assumptions['custom_investment_rate'] / 100);

    // Year-by-year projection
    for ($age = $currentAge; $age < $deathAge; $age++) {
        // Step 1: Calculate cash surplus for this year
        if ($age < $retirementAge) {
            $income = $this->getTotalAnnualIncome($user);
            $expenses = $this->getCurrentAnnualExpenses($user);
        } else {
            $income = $this->getRetirementIncome($user, $age);
            $expenses = $this->getRetirementExpenses($user);
        }
        $surplus = $income - $expenses;

        // Step 2: Update cash balance
        $cashBalance += $surplus;

        // Step 3: If cash goes negative, draw from investments
        if ($cashBalance < 0) {
            $deficit = abs($cashBalance);
            $cashBalance = 0; // Reset cash to zero

            // Distribute deficit equally across all investment accounts
            $accountCount = count($investments);
            if ($accountCount > 0 && $totalInvestments > 0) {
                $deficitPerAccount = $deficit / $accountCount;

                foreach ($investments as &$account) {
                    $account['balance'] = max(0, $account['balance'] -
$deficitPerAccount);
                }
                unset($account);

                // Recalculate total after drawdown
                $totalInvestments = array_sum(array_column($investments,
'balance'));
            }
        }

        // Step 4: Apply investment growth AFTER drawdown
        foreach ($investments as &$account) {
            $account['balance'] *= (1 + $investmentGrowthRate);
        }
        unset($account);
        $totalInvestments = array_sum(array_column($investments, 'balance'));
    }

    return [
        'projected_cash' => round($cashBalance, 2),
        'projected_investments' => round($totalInvestments, 2),
        'investment_accounts' => $investments, // For individual account
    ];

```

```
projections
    ];
}

/**
 * Get all investment accounts with current balances
 */
private function getInvestmentAccounts(User $user, ?User $spouse, bool
$dataSharingEnabled): array
{
    $accounts = [];

    // User's investment accounts
    foreach ($user->investmentAccounts as $account) {
        if (!$account->is_iht_exempt) {
            $accounts[] = [
                'id' => $account->id,
                'name' => $account->account_name,
                'owner' => 'user',
                'balance' => (float) $account->current_value,
            ];
        }
    }

    // Spouse's investment accounts (if data sharing enabled)
    if ($dataSharingEnabled && $spouse) {
        foreach ($spouse->investmentAccounts as $account) {
            if (!$account->is_iht_exempt) {
                $accounts[] = [
                    'id' => $account->id,
                    'name' => $account->account_name,
                    'owner' => 'spouse',
                    'balance' => (float) $account->current_value,
                ];
            }
        }
    }

    return $accounts;
}
```

Integrated Projection Timeline Example (Mitchell Family):

Year	Age	Phase	Income	Expenses	Surplus	Cash	
Investments							
1	49	Pre-Ret	£265,000	£185,500	+£79,500	£193,730	£305,000
...							
11	59	Pre-Ret	£265,000	£185,500	+£79,500	£988,730	£611,000
12	60	Retired	£75,000	£135,000	-£60,000	£928,730	£639,000
...							
27	75	Retired	£75,000	£135,000	-£60,000	£28,730	

£1,200,000						
28	76	Retired	£75,000	£135,000	-£60,000	£0
£1,168,730		← Deficit drawn				
29	77	Retired	£75,000	£135,000	-£60,000	£0
£1,138,xxx		← £60k drawn				
...						
37	85	Retired	£75,000	£135,000	-£60,000	£0
						£xxx,xxx

Key Points:

- Cash cannot go negative; deficits draw from investments
- Investment growth is applied AFTER the annual drawdown
- Deficit is split equally across all accounts (fair distribution)
- If investments are exhausted, a warning is generated

7.4 Combined Projection Calculation

The combined projection uses the **integrated cash-investment model** to ensure that:

1. Cash deficits are drawn from investments before growth is applied
2. Properties and chattels are projected independently
3. Individual investment account values are tracked for the breakdown display

```
public function calculateProjectedValues(
    User $user,
    ?User $spouse,
    bool $isMarried,
    bool $dataSharingEnabled = false
): array {
    $currentAge = $user->date_of_birth->age ?? 50;
    $retirementAge = $this->getRetirementAge($user);
    $deathAge = $this->getEstimatedDeathAge($user, $spouse, $isMarried);
    $yearsToProject = $deathAge - $currentAge;

    // Use INTEGRATED cash-investment projection (cash deficits draw from
    investments)
    $cashInvestmentResult = $this->projectCashAndInvestmentsIntegrated(
        $user,
        $spouse,
        $currentAge,
        $retirementAge,
        $deathAge,
        $dataSharingEnabled
    );

    $projectedCash = $cashInvestmentResult['projected_cash'];
    $projectedInvestments = $cashInvestmentResult['projected_investments'];
    $investmentAccountBreakdown =
    $cashInvestmentResult['investment_accounts'];

    // Project properties independently (no cash interaction)
    $projectedProperties = $this->projectProperties($user, $yearsToProject);
```

```

    // Get chattels at current value (no appreciation assumed)
    $projectedChattels = $this->getCurrentChattelsValue($user, $spouse,
$dataSharingEnabled);

    // Project liabilities (mortgages, debts)
    $projectedLiabilities = $this->projectLiabilities(
        $user,
        $currentAge,
        $retirementAge,
        $deathAge
    );

    // Calculate totals (include chattels)
    $projectedGrossAssets = $projectedCash + $projectedInvestments +
    $projectedProperties + $projectedChattels;
    $projectedNetEstate = $projectedGrossAssets - $projectedLiabilities;

    // Calculate projected IHT
    $totalAllowances = $this->calculateAllowances($user, $spouse,
    $isMarried);
    $projectedTaxableEstate = max(0, $projectedNetEstate - $totalAllowances);
    $ihtRate = $this->determineIHTRate($user, $projectedNetEstate,
    $totalAllowances['nrb']);
    $projectedIHTLiability = $projectedTaxableEstate * $ihtRate['rate'];

    return [
        'projected_cash' => round($projectedCash, 2),
        'projected_investments' => round($projectedInvestments, 2),
        'projected_properties' => round($projectedProperties, 2),
        'projected_gross_assets' => round($projectedGrossAssets, 2),
        'projected_liabilities' => round($projectedLiabilities, 2),
        'projected_net_estate' => round($projectedNetEstate, 2),
        'projected_taxable_estate' => round($projectedTaxableEstate, 2),
        'projected_iht_liability' => round($projectedIHTLiability, 2),
        'years_to_death' => $yearsToProject,
        'retirement_age' => $retirementAge,
        'estimated_age_at_death' => $deathAge,
    ];
}

```

7.5 Assumptions Summary

Asset Type	Projection Method	Default Value	Configurable
Cash Accounts	Income - Expenses surplus	n/a	No (derived)
Investment Accounts	Monte Carlo (p20) with drawdown	80% confidence	Yes
Properties	Compound growth	3% annual	Yes
Chattels/Valuables	No growth (current value)	0%	No
Mortgages	Linear amortisation	End at retirement	No
Other Liabilities	Linear amortisation	End at retirement	No

Asset Type	Projection Method	Default Value	Configurable
Inflation	Applied to expenses	2.5%	Yes
State Pension	Included from SPA	Actual amount	No

Integrated Cash-Investment Model:

- Cash and investments are projected together year-by-year
- When cash goes negative, deficit is drawn from investments BEFORE growth
- Deficit is split equally across all investment accounts
- Growth is then applied to the reduced investment balance

7.6 Key Transition Points

Timeline: Current Age → Retirement Age → State Pension Age → Death

Pre-Retirement Phase:

- | Income: Employment, self-employment, rental, dividends, interest, other
- | Expenses: `ExpenditureProfile.total_monthly_expenditure × 12` (or income × 70%)
- | Cash: Grows by (Income – Expenses) annually
- | Investments: Grow at Monte Carlo rate (no drawdown typically needed)

Post-Retirement Phase:

- | Income: `target_retirement_income + pension drawdown`
- | State Pension: Added from `state_pension_age` onwards
- | Expenses: essential + lifestyle (or `target_retirement_income` or income × 50%)
- | Cash: Grows/shrinks by (Retirement Income – Retirement Expenses)
- | Investments: IF cash goes negative → drawdown deficit → THEN apply growth

Cash-Investment Interaction:

- | IF cash surplus positive → cash grows, investments grow independently
- | IF cash surplus negative AND cash > 0 → cash reduces, investments grow
- | IF cash surplus negative AND cash = 0 → deficit drawn from investments
- | IF investments exhausted → estate insolvency warning

Liability End Points:

- | Mortgages: End at `mortgage.end_date` OR `retirement_age`
- | Other Debts: End at `liability.end_date` OR `retirement_age`

7.7 Edge Cases

Scenario	Handling
No income data	Use £0 (conservative - no cash growth)
No expenditure profile	Use £0 (all income becomes surplus)
No retirement profile	Use 50% of pre-retirement income as retirement expenses

Scenario	Handling
Already retired	Skip pre-retirement phase, use retirement figures
Negative cash surplus	Draw deficit from investment accounts equally
Cash goes to zero	All future deficits drawn from investments
Investments exhausted	Generate warning; estate likely insolvent before death
No end date on liability	Assume cleared at retirement age
Already past retirement age	All liabilities assume £0 remaining
No investment accounts	Cash deficit cannot be covered; cap cash at £0
Single investment account	All deficit drawn from that one account

Investment Drawdown Sequence:

1. Cash balance goes negative → deficit calculated
2. Deficit divided equally by number of investment accounts
3. Each account reduced by `deficit ÷ account_count`
4. Investment growth applied to reduced balance
5. Process repeats for each remaining year

Warning Conditions:

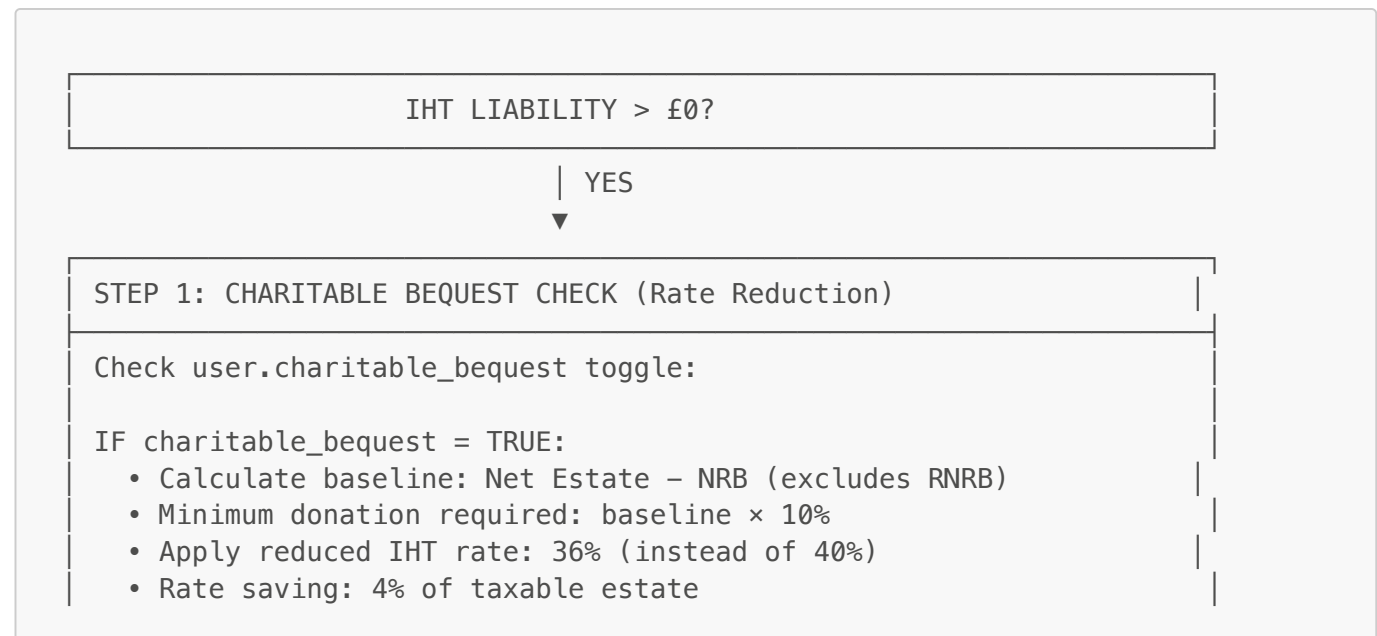
- If total investments fall below £0 before death age → "Estate may be insolvent"
- If projected cash + investments < 0 → "Lifestyle unsustainable without asset sales"

8. IHT Mitigation Strategies

8.1 Decision Tree

File: `EstateAgent.php` lines 125-204

The mitigation strategy follows a **cost-efficient priority order**, using CLTs only as a last resort:



IF charitable_bequest = FALSE:

- Use standard IHT rate: 40%
- Consider recommending charitable giving if:
 - User has existing charitable intentions
 - Rate saving exceeds emotional cost of bequest

➤ effective_iht_rate = charitable_bequest ? 0.36 : 0.40

➤ Recalculate IHT liability using effective rate

Data Sources:

- Toggle: users.charitable_bequest (boolean)
- Percentage: iht_profiles.charitable_giving_percent (decimal)
- Rate: TaxConfigService.get('inheritance_tax.reduced_rate_charity')



STEP 2: LIQUIDITY & AFFORDABILITY ASSESSMENT

- Query all assets from EstateAssetAggregatorService
- Categorise: liquid (cash, investments) vs illiquid (property)
- Calculate surplus income (income – expenditure)
- Identify affordability constraints for ongoing commitments



STEP 3: CHECK EXISTING LIFE COVER

- Query Protection module for all life policies
- Check: Is policy written in trust? (bypasses estate if yes)
- Calculate: Usable Cover = Total Cover – Outstanding Debts
- Only excess above debts can offset IHT liability



STEP 4: ANNUAL GIFTING STRATEGY (First Resort)

Immediately exempt gifts – no 7-year wait, no tax risk:

- Annual Exemption: £3,000/year × years to life expectancy
- Normal Expenditure from Income: surplus × safe percentage
- Small Gifts: £250 per recipient (unlimited recipients)
- Wedding Gifts: £5,000 (parent), £2,500 (grandparent), £1,000

➤ IF total covers IHT liability → RECOMMEND (stop here)

➤ IF NOT → remaining_liability = IHT – gifting_capacity



STEP 5: LIFE COVER STRATEGY (Second Resort)

IF user age ≤ 50 :

- Calculate whole of life premium for remaining liability
- Check premium affordability against surplus income
- Policy MUST be written in trust to bypass estate
- RECOMMEND if affordable

IF user age > 50 :

- Life cover premiums likely prohibitively expensive
- Only use EXISTING cover excess (Step 3)
- Skip new life cover → proceed to PET strategy

► remaining_liability = IHT - gifting - life_cover_excess



STEP 6: PET GIFTING STRATEGY (Third Resort)

Potentially Exempt Transfers – exempt if donor survives 7 years:

- Calculate 7-year cycles: $\text{floor}(\text{years_to_death} / 7)$
- Each cycle can gift up to NRB (£325,000) tax-efficiently
- Apply taper relief if donor likely to die within 7 years

IF user age > 50 AND no existing life cover:
→ PRIORITISE PETs over purchasing new life cover

► remaining_liability = IHT - gifting - life_cover - PETs



STEP 7: CLT INTO TRUST (Last Resort ONLY)

⚠ Only if Steps 4–6 do NOT fully cover the liability

Chargeable Lifetime Transfer costs:

- Immediate 20% charge on amount exceeding available NRB
- Additional 20% if death within 7 years (40% total)
- Periodic charges: max 6% every 10 years
- Exit charges: proportionate to time since last anniversary

RECOMMEND with:

- Full cost/benefit analysis
- Comparison of immediate charge vs potential IHT saving
- Multi-cycle strategy if time permits (7-year cycles)

Additional Recommendations (if health score < 50):

- └ Complete IHT profile
- └ Review beneficiary designations
- └ Write/update will

The 36% reduced rate requires leaving at least 10% of the "baseline amount" to charity:

```
// Baseline calculation (excludes RNRB - only NRB counts)
$baseline = max(0, $netEstate - $nrbAvailable);

// Minimum charitable donation for reduced rate
$minimumDonation = $baseline * 0.10;

// IHT rate determination
$ihtRate = ($charitableDonation >= $minimumDonation) ? 0.36 : 0.40;

// Potential saving from charitable giving
$potentialSaving = $taxableEstate * 0.04; // 40% - 36% = 4% saving
```

Example:

- Net Estate: £1,000,000
- NRB Available: £325,000
- Baseline: £675,000
- Minimum Donation: £67,500 (10% of baseline)
- Taxable Estate: £675,000
- IHT at 40%: £270,000
- IHT at 36%: £243,000
- Saving: £27,000 (by donating £67,500 to charity)

Charity Code Consolidation (Implementation Note)

The charity logic currently exists in multiple locations and needs consolidation:

Location	Current State	Consolidation Action
<code>users.charitable_bequest</code>	Boolean toggle	Keep as user preference
<code>iht_profiles.charitable_giving_percent</code>	Decimal percentage	Consolidate - use this as source of truth
<code>IHTCalculator.calculateCharitableReduction()</code>	Returns 0.36 or 0.40	Keep - uses TaxConfigService
<code>IHTCalculationService</code>	Does NOT use <code>charitable_giving_percent</code>	FIX - integrate charity rate
<code>IHTPlanning.vue</code>	Frontend calculations only	Keep for display, backend is source of truth

Location	Current State	Consolidation Action
<code>GiftingStrategy.php</code>	Recommends 10% giving	Keep as recommendation engine

Required Changes:

1. **IHTCalculationService.php** - Must query `IHTProfile.charitable_giving_percent` and apply reduced rate:

```
// In calculateIHT() method
$charitablePercent = $ihtProfile->charitable_giving_percent ?? 0;
$ihtRate = $this->ihtCalculator->calculateCharitableReduction(
    $netEstate,
    $charitablePercent
);
```

2. **Sync Toggle and Percentage** - When `user.charitable_bequest` is toggled:

- TRUE → Set `iht_profile.charitable_giving_percent = 10` (minimum)
- FALSE → Set `iht_profile.charitable_giving_percent = 0`

3. **Single Source of Truth** - All calculations should use:

- Rate: `TaxConfigService.get('inheritance_tax.reduced_rate_charity')` → 0.36
- Threshold: `TaxConfigService.get('inheritance_tax.charity_threshold_percent')` → 0.10
- Never hardcode 0.36 or 10% in services

8.2 Gifting Strategies

File: `app/Services/Estate/GiftingStrategyOptimizer.php`

Priority 1: Annual Exemption

```
// £3,000 per year, immediately exempt
$totalGifted = 3000 * $yearsUntilDeath;
$ihtSaved = $totalGifted * 0.40;

// Example: 20 years × £3,000 = £60,000 gifted
// IHT saved: £60,000 × 40% = £24,000
```

Priority 2: Normal Expenditure from Income

```
// Must be regular, from surplus income, not affecting standard of living
$surplusIncome = $totalIncome - $annualExpenditure;
```

```
$safeGiftingAmount = $surplusIncome * 0.5; // Conservative 50%

// Minimum £1,000/year to be worthwhile
// Immediately exempt if regular pattern established (3+ years)
```

Priority 3: Potentially Exempt Transfers (PETs)

```
// 7-year cycle strategy
$complete7YearCycles = floor($yearsUntilDeath / 7);
$amountPerCycle = $totalNRBAvailable; // Up to NRB per cycle

// Gift becomes exempt after 7 years
// Taper relief applies 3-7 years
```

Years	Tax Rate
0-3	40%
3-4	32%
4-5	24%
5-6	16%
6-7	8%
7+	0%

Priority 4: Chargeable Lifetime Transfers (CLTs)

```
// Last resort - attracts immediate charge
$immediateCLTCharge = $targetGiftAmount * 0.20; // 20% on excess over NRB

// If death within 7 years: additional charge to bring total to 40%
$potentialDeathCharge = ($excessOverNRB * 0.40) - $immediateCLTCharge;
```

8.3 Trust Strategies

File: `app/Services/Estate/PersonalizedTrustStrategyService.php`

Strategy 1: Immediate Discretionary Trust (CLT)

```
[
    'strategy_name' => 'Immediate Discretionary Trust (CLT)',
    'priority' => 1,
    'amount' => $liquidAssets,
    'iht_saving_potential' => min($amount, $availableNRB) * 0.40,
    'lifetime_tax_charge' => max(0, $amount - $availableNRB) * 0.20,
```

```
'time_frame' => '7 years for full effectiveness',
'risk_level' => $excessOverNRB > 0 ? 'Medium' : 'Low',
]
```

Benefits:

- Assets immediately outside estate
- Growth outside estate
- Flexibility for trustees
- Protection from creditors

Risks:

- 20% immediate charge if over NRB
- Additional 20% if death within 7 years
- 10-year anniversary charges (6%)
- Loss of control

Strategy 2: Multi-Cycle CLT

```
// Use multiple 7-year cycles
$cyclesPossible = floor($yearsUntilDeath / 7);
$amountPerCycle = $availableNRB;
$totalOverLifetime = $amountPerCycle * $cyclesPossible;
```

Example:

- 21 years until death = 3 cycles
- £325,000 per cycle = £975,000 total
- No immediate charge (each within NRB)
- IHT saved: £975,000 × 40% = £390,000

Strategy 3: Loan Trust

```
// Loan to trust (NOT a gift)
$loanAmount = $liquidAssets;
$assumedGrowthRate = 0.05; // 5%/year
$growthOver20Years = $loanAmount * (pow(1.05, 20) - 1);
$ihtSaving = $growthOver20Years * 0.40;
```

Benefits:

- No immediate IHT charge
- Growth outside estate
- Retain access to capital
- No 7-year wait for growth benefit

Mechanics:

- Loan stays in estate (no IHT benefit on principal)
- Investment growth is outside estate
- Can write off loan using annual exemptions (£3,000/year)

Strategy 4: Discounted Gift Trust

```
// Gift with retained income rights
$discountRate = min(0.50, ($incomeRate * min($lifeExpectancy, 20)) / 2);
$giftValue = $totalLiquid;
$discountValue = $giftValue * $discountRate;
$cltValue = $giftValue - $discountValue; // Lower chargeable amount
```

Benefits:

- Reduced CLT value
- Retain income for life
- All growth outside estate

Strategy 5: Property Trust Planning

Cannot gift main residence and continue living in it (GROB rules)

Options:

- **Downsizing:** Release equity for gifting
- **Life Interest Trust (in Will):** Spouse lives in property for life
- **Shared Ownership:** Gift % to children, they pay market rent

8.4 Life Insurance Strategy

File: `ComprehensiveEstatePlanService.php`

```
// Whole of Life policy in trust
[
    'recommended_approach' => 'Whole of Life Policy',
    'sum_assured_required' => $ihtLiability,
    'written_in_trust' => true, // Proceeds outside estate
]
```

Premium Estimation:

```
$monthlyRatePer1000 = 0.50 + ($age - 40) * 0.05;
$monthlyPremium = ($sumAssured / 1000) * $monthlyRatePer1000;
```

9. Will Integration

9.1 Will Model

File: `app/Models/Estate/Will.php`

```
[
    'has_will' => true,
    'spouse_primary_beneficiary' => true,
    'spouse_bequest_percentage' => 100,
    'executor_name' => 'John Smith',
    'executor_notes' => 'Professional executor',
    'will_last_updated' => '2024-01-15',
]
```

9.2 Bequest Types

Type	Description
<code>percentage_of_estate</code>	e.g., "50% of residual estate"
<code>specific_amount</code>	e.g., "£10,000 to charity"
<code>specific_asset</code>	e.g., "Main residence to eldest child"

9.3 Intestacy Rules

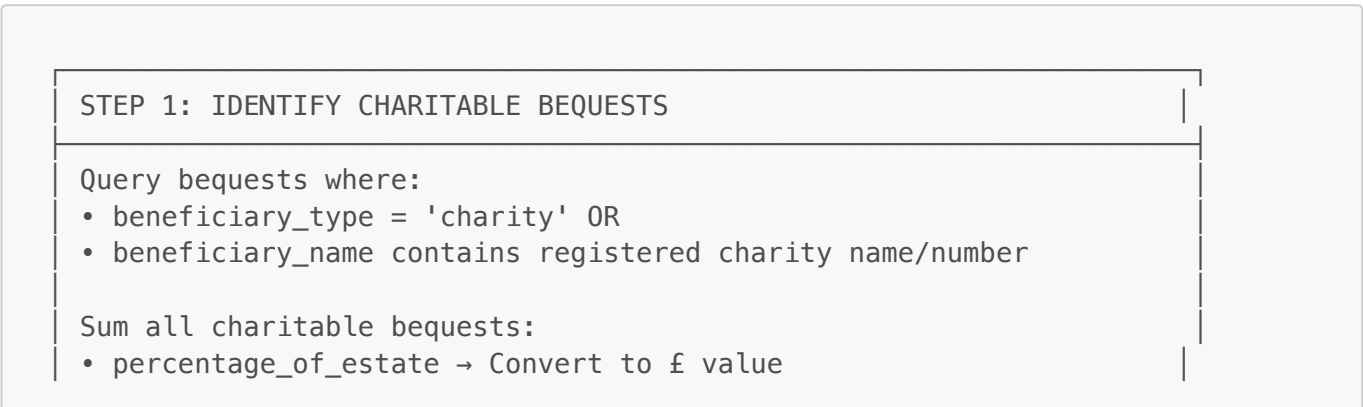
File: `resources/js/components/Estate/IntestacyRules.vue`

Scenario	Distribution
Spouse + children	Spouse: £322,000 + 50% remainder; Children: 50% remainder
Children only	Equal shares
Spouse only	100% to spouse
No relatives	Escheated to Crown

9.4 Charitable Bequest Analysis

The system automatically checks charitable bequests against the 10% threshold for the reduced 36% IHT rate.

Bequest Evaluation Flow



- `specific_amount` → Direct value
- `specific_asset` → Use `asset_current_value`



STEP 2: CALCULATE THRESHOLD

`baseline` = Net Estate – NRB (excludes RNRB)
`threshold` = `baseline` × 10%

Example: £1,000,000 net estate – £325,000 NRB = £675,000 baseline
 Threshold = £67,500



STEP 3: COMPARE AND INFORM

IF `charitable_bequest_total` < `threshold`:

- Status: BELOW THRESHOLD
- IHT Rate: 40% (standard)
- Shortfall: `threshold` – `charitable_bequest_total`
- Message: "Increase charitable bequests by £X to qualify for 36% reduced rate and save £Y in IHT"

IF `charitable_bequest_total` = `threshold` (±1%):

- Status: AT THRESHOLD
- IHT Rate: 36% (reduced)
- Message: "Your charitable bequests qualify for the reduced 36% IHT rate. Consider increasing slightly for margin of safety."

IF `charitable_bequest_total` > `threshold`:

- Status: ABOVE THRESHOLD
- IHT Rate: 36% (reduced)
- Excess: `charitable_bequest_total` – `threshold`
- Message: "Charitable bequests exceed minimum threshold. You comfortably qualify for the 36% reduced rate."



STEP 4: AUTO-ADJUST IHT CALCULATION

IF `status` = `AT_THRESHOLD` or `ABOVE_THRESHOLD`:

- Set `effective_iht_rate` = 0.36
- Recalculate: `iht_liability` = `taxable_estate` × 0.36
- Update `IHTProfile.charitable_giving_percent`
- Sync with `user.charitable_bequest` toggle

IF `status` = `BELOW_THRESHOLD`:

- Set `effective_iht_rate` = 0.40
- Generate recommendation to increase charitable giving

- Show potential IHT saving if threshold met

Charitable Bequest Status Display

```
// In WillAnalysisService or IHTCalculationService
public function analyzeCharitableBequests(User $user, float $netEstate):
array
{
    $nrb = $this->taxConfig->get('inheritance_tax.nil_rate_band');
    $baseline = max(0, $netEstate - $nrb);
    $threshold = $baseline * 0.10;

    $charitableBequests = $this->getCharitableBequestTotal($user);

    if ($charitableBequests >= $threshold) {
        $status = $charitableBequests > $threshold * 1.01 ? 'above' : 'at';
        $effectiveRate = 0.36;
        $shortfall = 0;
        $excess = $charitableBequests - $threshold;
    } else {
        $status = 'below';
        $effectiveRate = 0.40;
        $shortfall = $threshold - $charitableBequests;
        $excess = 0;
    }

    $potentialSaving = $baseline * 0.04; // 40% - 36% = 4%

    return [
        'status' => $status, // 'below', 'at', 'above'
        'charitable_total' => $charitableBequests,
        'threshold' => $threshold,
        'shortfall' => $shortfall,
        'excess' => $excess,
        'effective_rate' => $effectiveRate,
        'potential_saving' => $status === 'below' ? $potentialSaving : 0,
        'current_saving' => $status !== 'below' ? $potentialSaving : 0,
    ];
}
```

9.5 Trust-Triggering Wishes

Certain wishes in a will indicate the need for trust structures to be created. The system scans bequest notes and executor instructions for these patterns.

Wish Patterns That Trigger Trust Recommendations

Wish Pattern	Trust Type Suggested	Reason
--------------	----------------------	--------

Wish Pattern	Trust Type Suggested	Reason
"Education provided for children/kids"	Bare Trust or 18-25 Trust	Funds held until child reaches education age
"Income provided for family/spouse"	Interest in Possession Trust	Provides income stream while preserving capital
"Income provided for children"	Discretionary Trust	Flexible distribution for minors
"Inheritance at age 25"	Age 18-25 Trust	Delays full access until specified age
"Protect from divorce/creditors"	Discretionary Trust	Asset protection
"Special needs provision"	Disabled Person's Trust	IHT-efficient, preserves benefits
"Business to continue"	Business Property Trust	Succession planning
"Property to be managed"	Property Trust	Professional management

Wish Detection Logic

```
// In WillAnalysisService
public function detectTrustTriggeringWishes(Will $will): array
{
    $triggers = [];
    $wishPatterns = [
        'education_trust' => [
            'patterns' => ['education', 'school fees', 'university',
'college'],
            'trust_type' => 'bare_trust',
            'description' => 'Education Trust for Children',
            'iht_treatment' => 'Bare trust = PET, not CLT',
        ],
        'income_family' => [
            'patterns' => ['income for family', 'income for spouse', 'living
expenses'],
            'trust_type' => 'interest_in_possession',
            'description' => 'Interest in Possession Trust',
            'iht_treatment' => 'Pre-2006 IIP = not relevant property',
        ],
        'income_children' => [
            'patterns' => ['income for child', 'income for kids',
'maintenance'],
            'trust_type' => 'discretionary',
            'description' => 'Discretionary Trust for Minors',
            'iht_treatment' => 'Relevant property – 10-year charges apply',
        ],
        'age_restriction' => [
            'patterns' => ['at age 25', 'when they reach', 'upon turning'],
            'trust_type' => 'age_18_to_25',
            'description' => 'Age 18-25 Trust',
            'iht_treatment' => 'Special treatment – reduced exit charges',
        ],
        'asset_protection' => [
```

```

        'patterns' => ['protect from divorce', 'creditor protection',
'bankruptcy'],
        'trust_type' => 'discretionary',
        'description' => 'Asset Protection Trust',
        'iht_treatment' => 'Relevant property – full charges apply',
    ],
    'special_needs' => [
        'patterns' => ['special needs', 'disability', 'disabled',
'vulnerable'],
        'trust_type' => 'disabled_person',
        'description' => 'Disabled Person\'s Trust',
        'iht_treatment' => 'Exempt from periodic/exit charges',
    ],
];

// Scan bequest notes and executor instructions
$textToScan = collect($will->bequests)->pluck('notes')->join(' ')
    . ' ' . ($will->executor_notes ?? '');

foreach ($wishPatterns as $key => $config) {
    foreach ($config['patterns'] as $pattern) {
        if (stripos($textToScan, $pattern) !== false) {
            $triggers[] = [
                'wish_type' => $key,
                'matched_pattern' => $pattern,
                'trust_type' => $config['trust_type'],
                'description' => $config['description'],
                'iht_treatment' => $config['iht_treatment'],
                'recommendation' => "Consider creating a
{$config['description']} to fulfil this wish",
            ];
            break; // One match per category is enough
        }
    }
}

return $triggers;
}

```

Integration with Estate Planning

When trust-triggering wishes are detected:

1. **Alert in Will Summary** - Display detected wishes with trust recommendations
2. **Link to Trust Strategy** - Connect with [PersonalizedTrustStrategyService](#) for detailed recommendations
3. **IHT Impact Analysis** - Show how the suggested trust would affect IHT:
 - Entry charge (if CLT)
 - Periodic charges (if relevant property)
 - Potential IHT savings

```
// In EstateAgent.php – integrate wish detection
public function analyze($userId): array
{
    // ... existing analysis ...

    // Check will for trust-triggering wishes
    $will = Will::where('user_id', $userId)->with('bequests')->first();
    if ($will) {
        $wishTriggers = $this->willAnalysisService-
        >detectTrustTriggeringWishes($will);

        if (!empty($wishTriggers)) {
            $analysisData['trust_wish_triggers'] = $wishTriggers;
            $analysisData['recommendations'][] = [
                'type' => 'will_trust_setup',
                'priority' => 'medium',
                'title' => 'Will Wishes Require Trust Structures',
                'description' => count($wishTriggers) . ' wishes in your will
may require trust arrangements',
                'details' => $wishTriggers,
            ];
        }
    }

    return $analysisData;
}
```

9.6 Integration Points

- Will status affects estate health score
- Executor information in probate readiness
- Bequests inform asset distribution scenarios
- Asset beneficiary designations can override will
- Charitable bequests auto-adjust IHT rate calculation
- Wish patterns trigger trust recommendations

10. User Type Handling

10.1 Single Users

```
// Standard calculation
$nrpAvailable = 325000;           // Single NRB
$nrpAvailable = 175000;           // If main residence owned
$yearsUntilDeath = $this->calculateLifeExpectancy($user);
```

10.2 Married/Civil Partnership

```
// Combined calculation at second death
$isMarried = in_array($user->marital_status, ['married']) && $spouse !==
null;

if ($isMarried && $dataSharingEnabled) {
    // Combine assets
    $totalGrossAssets = $userGrossAssets + $spouseGrossAssets;
    $totalLiabilities = $userLiabilities + $spouseLiabilities;

    // Combined allowances
    $nrbAvailable = 650000; // £325k × 2
    $rnrAvailable = 350000; // £175k × 2 (if main residence)

    // Second death projection
    $yearsUntilDeath = max(
        $this->calculateLifeExpectancy($user),
        $this->calculateLifeExpectancy($spouse)
    );
}
```

Key Features:

- Spouse exemption: Transfers between spouses are IHT-free
- Transferable NRB: Unused spouse NRB transfers to survivor
- Transferable RNRB: Unused RNRB also transfers
- Second death scenario: Projects to longer-lived spouse

10.3 Widowed Users

```
// Check for transferred allowances
$transferredNRB = $ihtProfile->nrb_transferred_from_spouse ?? 0;

// Maximum: £650,000 (own £325k + transferred £325k)
$nrbAvailable = 325000 + min(325000, $transferredNRB);

// RNRB can also be enhanced
// Up to £500k if both main residence benefits available
```

10.4 Divorced Users

- Standard single logic applies
- No transferable allowances
- Can remarry for new spouse benefits

11. Linked Accounts (Spouse Integration)

11.1 Account Linking

```
// User model relationships
$user->spouse_id;           // Links to spouse User record
$user->spouse;              // BelongsTo relationship
$user->hasAcceptedSpousePermission(); // Data sharing consent
```

11.2 Data Sharing Permission

```
// Required for viewing spouse financial data
$dataSharingEnabled = $spouse && $user->hasAcceptedSpousePermission();

if ($dataSharingEnabled) {
    // Fetch spouse assets
    $spouseAssets = $this->assetAggregator->gatherUserAssets($spouse);
    // Calculate combined IHT
    // Show dual balance sheets
}
```

11.3 Joint Asset Pattern

```
// Single record, ownership split by percentage
Property::create([
    'user_id' => $user->id,
    'joint_owner_id' => $spouse->id,
    'ownership_type' => 'tenants_in_common',
    'ownership_percentage' => 70, // User gets 70%, spouse gets 30%
    'current_value' => 320000,
]);

// Query for either owner
$properties = Property::where('user_id', $user->id)
    ->orWhere('joint_owner_id', $user->id)
    ->get();

// Calculate share
$userShare = $property->user_id === $user->id
    ? $property->current_value * ($property->ownership_percentage / 100)
    : $property->current_value * ((100 - $property->ownership_percentage) /
100);
```

11.4 Dual Balance Sheets

File: `ComprehensiveEstatePlanService.php` lines 575-852

```
// When data sharing enabled
[
    'user' => [
        'name' => 'John Smith',
```

```
      'total_assets' => 500000,
      'total_liabilities' => 100000,
      'net_worth' => 400000,
    ],
    'spouse' => [
      'name' => 'Jane Smith',
      'total_assets' => 300000,
      'total_liabilities' => 50000,
      'net_worth' => 250000,
    ],
    'combined' => [
      'total_assets' => 800000,
      'total_liabilities' => 150000,
      'net_worth' => 650000,
    ],
  ],
]
```

12. Frontend Architecture

12.1 Vuex Store

File: resources/js/store/modules/estate.js

State

```
const state = {
  assets: [],
  investmentAccounts: [],
  liabilities: [],
  gifts: [],
  trusts: [],
  ihtProfile: null,
  netWorth: null,
  cashFlow: null,
  analysis: null,
  recommendations: [],
  secondDeathPlanning: null,
  loading: false,
  error: null,
};
```

Key Getters

Getter	Description
allAssets	Combines manual + investment accounts
totalAssets	Sum of all asset values
totalLiabilities	Sum of all liabilities

Getter	Description
netWorthValue	Assets - Liabilities
ihlLiability	Current IHT liability
taxableEstate	Estate after allowances
grossEstate	Estate before allowances
giftsWithin7Years	PETs in 7-year window
futureDeathAge	Projected age at death
futureTaxableEstate	Projected taxable estate
futureIHTLiability	Projected IHT liability

Key Actions

Action	Description
fetchEstateData	Load all estate data
calculateIHT	Trigger IHT calculation
calculateSecondDeathIHTPlanning	Married couple planning
fetchNetWorth	Net worth analysis
fetchCashFlow	Cash flow projections
createAsset / updateAsset / deleteAsset	Asset CRUD
createLiability / updateLiability / deleteLiability	Liability CRUD
createGift / updateGift / deleteGift	Gift CRUD

12.2 Vue Components

Location: [resources/js/components/Estate/](#)

Planning Components

Component	Purpose
IHTPlanning.vue	IHT analysis and calculations
IHTMitigationStrategies.vue	Strategy recommendations
TrustPlanning.vue	Trust planning interface
WillPlanning.vue	Will creation/management
GiftingStrategy.vue	Gifting strategy UI
LifePolicyStrategy.vue	Life insurance recommendations

Data Management

Component	Purpose
AssetForm.vue	Add/edit assets
AssetsLiabilities.vue	Asset/liability list view
LiabilityForm.vue	Add/edit liabilities
GiftForm.vue	Add/edit gifts
TrustForm.vue	Add/edit trusts

Visualisation

Component	Purpose
IHTCalculationTable.vue	IHT breakdown table
IHTAssetBreakdown.vue	Asset category breakdown
IHTLiabilityGauge.vue	Visual gauge for IHT
EstateProjectionComparison.vue	Current vs projected
NRBRNRBTracker.vue	NRB/RNRB utilization
DualGiftingTimeline.vue	Gifting schedule
NetWorthWaterfallChart.vue	Waterfall visualization

13. API Endpoints

File: app/Http/Controllers/Api/EstateController.php

Core Endpoints

Method	Endpoint	Handler	Purpose
GET	/api/estate	index()	Fetch all estate data
GET	/api/estate/comprehensive-plan	getComprehensiveEstatePlan()	Full estate plan
GET	/api/estate/net-worth	getNetWorth()	Net worth analysis
GET	/api/estate/cash-flow	getCashFlow()	Cash flow projections

Asset Endpoints

Method	Endpoint	Handler
POST	/api/estate/assets	storeAsset()

Method	Endpoint	Handler
PATCH	/api/estate/assets/{id}	updateAsset()
DELETE	/api/estate/assets/{id}	destroyAsset()

Liability Endpoints

Method	Endpoint	Handler
POST	/api/estate/liabilities	storeLiability()
PATCH	/api/estate/liabilities/{id}	updateLiability()
DELETE	/api/estate/liabilities/{id}	destroyLiability()

Gift Endpoints

Method	Endpoint	Handler
POST	/api/estate/gifts	storeGift()
PATCH	/api/estate/gifts/{id}	updateGift()
DELETE	/api/estate/gifts/{id}	destroyGift()

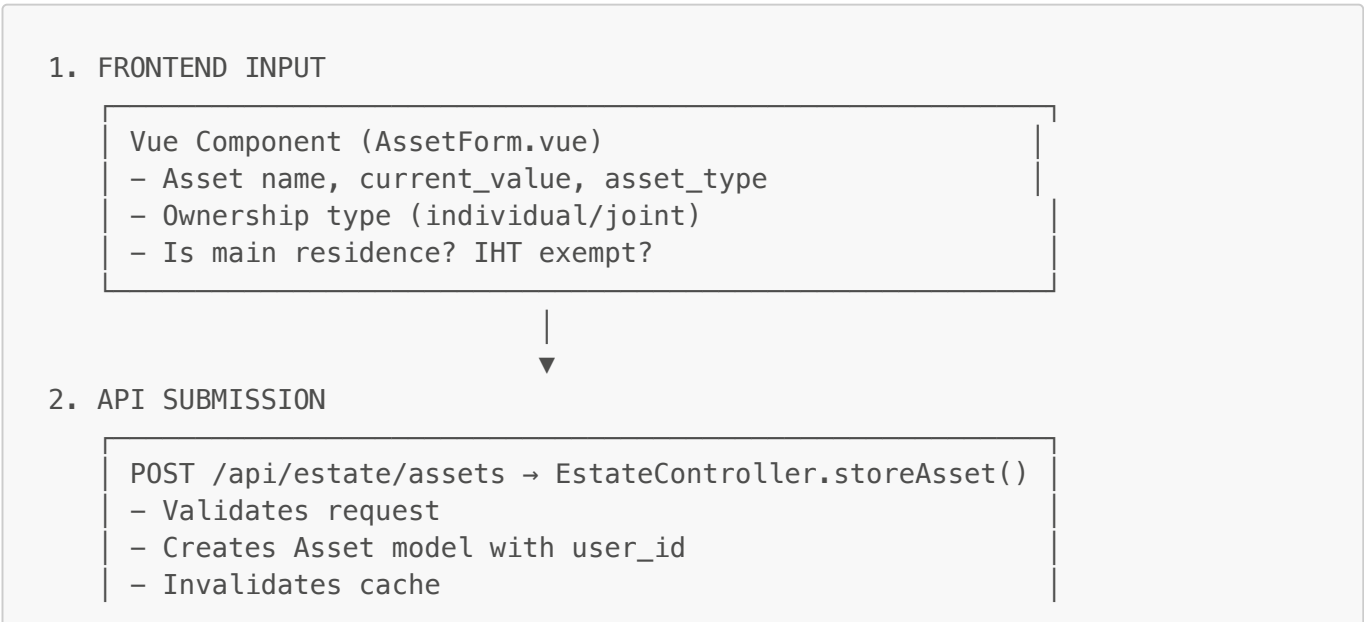
Cache Invalidation

All write operations invalidate cache:

```
Cache::forget("estate_analysis_{$user->id}");
```

14. Complete Data Flow

Example: User Creates Asset → IHT Recalculation → Display



- Returns created asset



3. STATE UPDATE

Vuex estate module

- Commit 'addAsset' mutation
- UI automatically updates via getters
- totalAssets getter recalculates



4. IHT CALCULATION

User clicks "Calculate IHT"

- `$store.dispatch('estate/calculateIHT')`
- `estateService.calculateIHT()`
- `POST /api/estate/calculate-ih`



5. BACKEND CALCULATION

`IHTCalculationService.calculate($user, $spouse, $flag)`

— EstateAssetAggregatorService.gatherUserAssets() —

- `Query Asset.where('user_id', $id)`
- Includes new asset
- Aggregates all module assets

— Calculate IHT —

- NRB: £325k (or £650k married)
- RNRB: Up to £175k (or £350k) if main residence
- Taxable = Net Estate - Allowances
- $IHT = Taxable \times 40\%$

— Project to Death —

- Life expectancy from actuarial tables
- Growth: $(1 + 4.7\%)^{\text{years}}$
- Projected IHT at death

- Save to IHTCalculation table
- Return full calculation



6. RECOMMENDATIONS (Optional)

`EstateAgent.generateRecommendations()`

- If > £100k: Trust + gifting recommendations
- `PersonalizedTrustStrategyService`: 5 trust strategies

- GiftingStrategyOptimizer: 4 gifting strategies
- Return prioritized recommendations



7. DISPLAY

- Vue Components render updated data:
- IHTCalculationTable: Breakdown
 - EstateProjectionComparison: Current vs projected
 - IHTMitigationStrategies: Recommended actions
 - TrustPlanning: Trust strategies with cost/benefit
 - GiftingStrategy: Gifting schedule

Appendix: File Reference

Backend Files

File	Purpose
app/Agents/EstateAgent.php	Module orchestrator
app/Services/Estate/IHTCalculationService.php	IHT calculations
app/Services/Estate/EstateAssetAggregatorService.php	Asset aggregation
app/Services/Estate/GiftingStrategyOptimizer.php	Gifting strategies
app/Services/Estate/PersonalizedTrustStrategyService.php	Trust strategies
app/Services/Estate/ComprehensiveEstatePlanService.php	Full plan generation
app/Services/Estate/AssetLiquidityAnalyzer.php	Liquidity analysis
app/Services/Estate/NetWorthAnalyzer.php	Net worth calculations
app/Services/TaxConfigService.php	Tax configuration
app/Http/Controllers/Api/EstateController.php	API controller
app/Models/Estate/*.php	All estate models

Frontend Files

File	Purpose
resources/js/store/modules/estate.js	Vuex store
resources/js/services/estateService.js	API service
resources/js/views/Estate/EstateDashboard.vue	Dashboard view
resources/js/views/Estate/ComprehensiveEstatePlan.vue	Full plan view
resources/js/components/Estate/*.vue	30+ components

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